BTS-V(SS)-08.15-0455 Reg.No.

D

B.Tech. Degree V Semester Special Supplementary Examination August 2015

ME 1506 POWER PLANT ENGINEERING

(2012 scheme)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Define the terms: (i) Maximum demand (ii) Demand factor (iii) Load factor (iv) Plant use factor (v) Diversity factor.

(b) Explain the functions of penstocks draft tubes and surge tanks with figures.

(c) What are the advantages and disadvantages of diesel engine power plants?

(d) Draw a two stage gas turbine cycle on a temperature entropy diagram and explain various processes.

(e) What are the mountings and accessories in a power plant boiler?

(f) Explain fluidized bed combustion.

(g) Explain the salient features of flat plate solar collector and evacuated tube solar collector with diagrams.

(h) Write a short note on fuel cells.

PART B

(4 × 15 = 60)

II. The daily demands of three consumers are given below. (15)

Time	Consumer 1	Consumer 2	Consumer 3
Midnight to 8 AM	No Load	200 W	No Load
8 AM to 2 PM	600 W	No Load	200 W
2 PM to 4 PM	200 W	1000 W	1200 W
4 PM to 10 PM	800 W	No Load	No Load
10 PM to Midnight	No Load	200 W	200 W

Plot the load curve and find (i) Maximum demand of individual consumers (ii) Load factor of individual consumers (iii) Diversity factor (iv) Load factor of the station.

OR

(P.T.O.)

III. A proposed plant has the following daily load cycle: (15)

Time in Hours	6-8	8-11	11-16	16-19	19-22	22-24	24-6
Load in MW	15	45	45	25	60	30	20

Draw the load curve and select suitable generating units from the following available, such as 8000 kW, 16000 kW, 20000 kW and 24000 kW. Up on selecting the generating units, calculate (i) Load factor and (ii) Plant use factor.

IV. Explain with neat sketches diesel engine power plant layout, and its components. (15)

OR

V. Explain the different types of combustion chamber design in gas turbine power plants. (15)

VI. Explain with sketches the different types of coal handling systems employed in steam power plants. (15)

OR

VII. How are the different types of nuclear reactor classified? Write briefly on each classification. (15)

VIII. (a) From Bets law find out the maximum percentage of power that can be extracted from wind. What are the other losses and the percentage of such power losses? (10)

(b) Define solar constant. What is the value of solar constant for earth? (5)

OR

IX. (a) Explain the principle of MHD power generation. (8)

(b) Explain the principle of OTEC system. (7)

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